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* Assignment no. 3 *

Q.1) A solar collector is a device that absorbs solar radiation and converts it into thermal energy (heat).

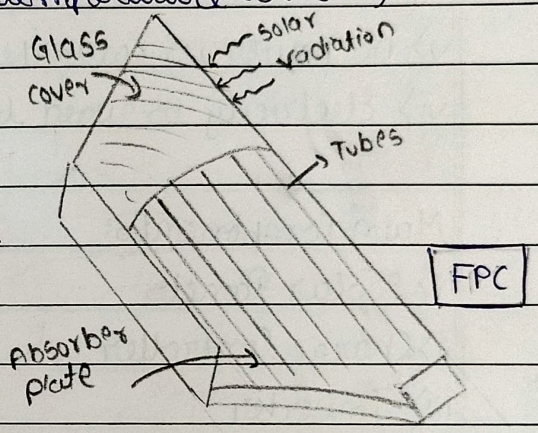
- Types of solar collector

i) Flat Plate Collector (FPC)

Features: works for low to medium temperature (30-80°C)

Components: Transparent glass cover,
Black absorbes plate,
Copper/aluminium tubes,
Insulation at the bottom.

Application: Solar water heaters,
space heating.



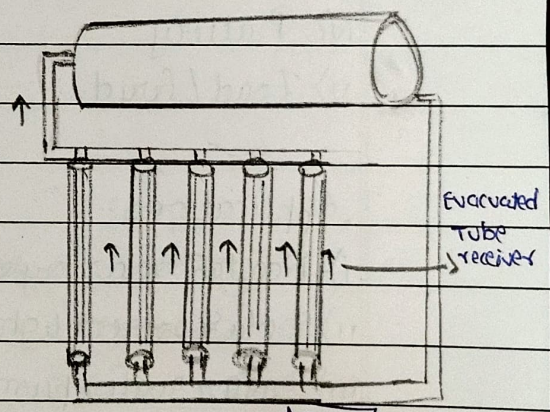
FPC

ii) Evacuated Tube Collector (ETC)

Feature: Very low heat loss,
suitable for colder regions.

Components: Double glass tubes with vacuum.
Absorber coating inside tube.

Applications: Domestic and Industrial
water heating.



ETC

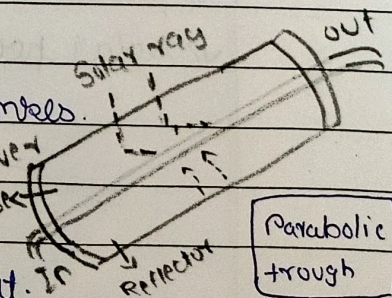
iii) Concentrating Solar Collectors:

Focus sunlight onto small receiver using mirror/lenses.

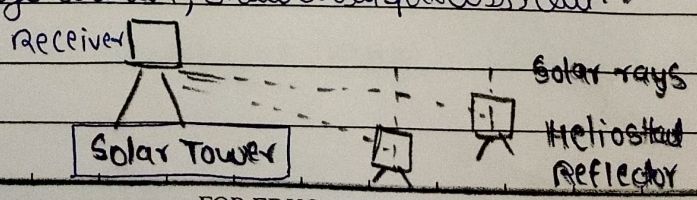
Types: Parabolic trough, solar dish, solar tower.

Features: Produces very high temperatures (>300°C)

Application: Power generation, Industrial process heat.



Parabolic trough



Solar Tower

Q.2) Working Principle of Solar PV System.

Solar PV system works on the photovoltaic effect.

i) Solar cells are made of semiconductor material (silicon)

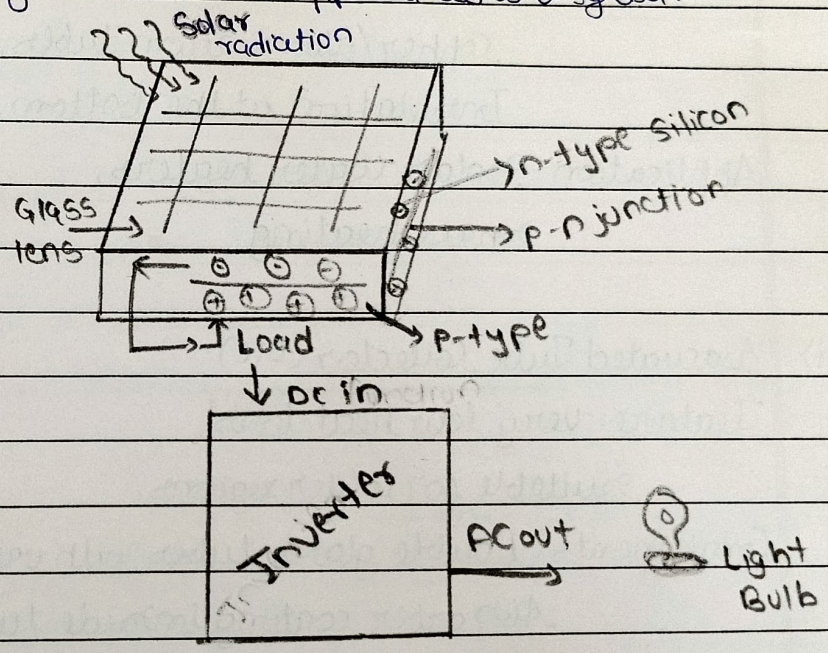
ii) Sunlight strikes the PV cell.

iii) Electrons get excited and start moving.

iv) This movement generates DC electricity.

v) DC power is converted into AC power using an inverter

vi) Electricity is used by loads or supplied to the grid.



Main Components:

i) Solar Panels

ii) Charge Controller

iii) Inverter

iv) Battery

v) Load / Grid.

Applications:

i) Rooftop solar system

ii) Solar street lighting

iii) Solar water pumps for agriculture

iv) Rural electrification

v) Solar power plants

Q.3) Major Solar Thermal Application:

i) Solar water Heating
- Used in homes, hotels, hospitals
- Saves electricity
- Reduces carbon emissions.

ii) Solar Cooking
- Solar cookers and community kitchens
- Reduces LPG consumption

iii) Solar Drying
- Drying agricultural products
- Improves food preservation

iv) Industrial Process Heat
- Textile, Dairy, food processing
- Low and medium temperature process.

v) Concentrated Solar Power (CSP)
- Electricity Generation
- Thermal storage enables night operation.

Advantages: High efficiency,
Clean and Renewable
Low operating cost
Reduces fossil fuel usage.

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